

1.7 Equilibria

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.7 Equilibria
Difficulty	Hard

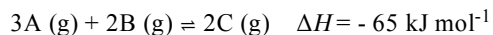
Time allowed: 80

Score: /60

Percentage: /100

Question 1a

The following dynamic equilibrium was reached at temperature, T , in a closed container.



The value of K_c for the reaction was $255 \text{ mol}^{-1} \text{ dm}^3$ when the equilibrium mixture contained 3.34 mol of **A** and 4.28 mol of **C**.

i)

Give the definition of dynamic equilibrium.

[2]

ii)

Write an expression for K_c for the reaction.

[1]

[3 marks]

Question 1b

Calculate the concentration of **B** in the equilibrium mixture if the volume of the container is 8.00 dm^3 . Give your answer to 3 significant figures.

[4 marks]

Question 1c

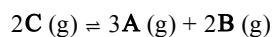
Changing the conditions of a closed container can have an effect on the concentrations of the reactants, products and K_c .

State the effect, if any, on the concentration of **B** at equilibrium if temperature, T , is increased and give a reason for your answer.

[2 marks]

Question 1d

Calculate the equilibrium constant for the following reaction at temperature, T .



[1 mark]

Question 2a

The graph in Fig. 2.1 shows the effect of pressure and temperature on the equilibrium yield of gaseous molecules.

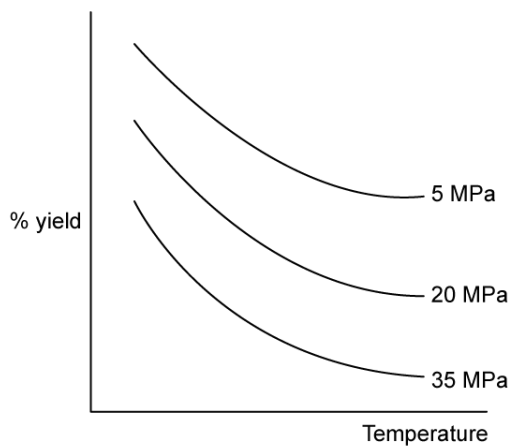


Fig. 2.1

Use Fig. 2.1 to **fully** explain whether the forward reaction is exothermic or endothermic.

[3 marks]

Question 2b

Use Fig 2.1 to explain whether the forward reaction will involve either an increase or decrease in the number of moles of a gas.

[3 marks]

Question 2c

The graph shows the relationship between temperature and K_c for a **different** dynamic equilibrium to produce a gaseous product is shown in Fig 2.2.

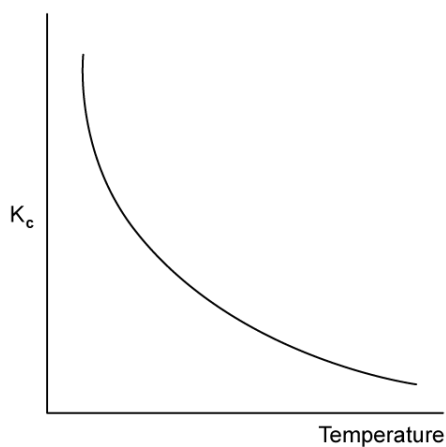


Fig 2.2

Use the information in Fig 2.2 to establish whether the **forward reaction** is exothermic or endothermic. Justify your answer.

[3 marks]

Question 2d

Gaseous products can be manufactured by the direct combination of reactants.

Explain why it is important for a chemist to know the value of K_c , at a given temperature, for such a reaction.

[2 marks]

Question 3a

A student carried out some acid-base titrations using different reactants.

The indicators and their pH range are given in Table 3.1.

Table 3.1

Indicator	pH range
methyl red	4.2 - 6.3
bromothymol blue	6.0 - 7.6
bromocresol green	3.8 - 5.4
phenolphthalein	8.2 - 10.0

In the first titration, they titrated 10.0 cm^3 of $0.200 \text{ mol dm}^{-3}$ HCl against $0.200 \text{ mol dm}^{-3}$ aqueous ammonia.

Sketch the pH curve that would be obtained from this titration.

[3 marks]

Question 3b

Using Table 3.1, explain why methyl red is a suitable indicator for the first titration but phenolphthalein is not.

[2 marks]

Question 3c

In a second titration, the student titrated 25.0 cm^3 of $0.200 \text{ mol dm}^{-3}$ sodium hydroxide against $0.100 \text{ mol dm}^{-3}$ ethanoic acid solution.

Sketch the graph that would be obtained for this titration.

[3 marks]

Question 3d

Using Table 3.1, explain why phenolphthalein is a suitable indicator for the second titration but methyl orange is not.

[2 marks]

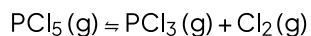
Question 3e

Explain whether the other two indicators in Table 3.1 are suitable for use in either titration.

[4 marks]

Question 4a

At high temperatures, gaseous phosphorus(V) chloride dissociates according to the following equation:



At 550K, the equilibrium amount of each gas is shown in Table 4.1.

Table 4.1

Chemical	$\text{PCl}_5(\text{g})$	$\text{PCl}_3(\text{g})$	$\text{Cl}_2(\text{g})$
Equilibrium moles	0.192	0.822	0.822

Calculate the total pressure of the reaction, if $K_p = 526 \text{ kPa}$. Give your answers to an appropriate number of significant figures and show your working.

$P = \dots\dots\dots \text{ kPa}$

[4 marks]

Question 4b

Nitrogen dioxide decomposes according to the following equation:



At 700 K, the equilibrium amount of each gas is shown in Table 4.2.

Table 4.2

Chemical	$\text{NO}_2(\text{g})$	$\text{NO}(\text{g})$	$\text{O}_2(\text{g})$
Equilibrium moles	0.92	0.06	0.04

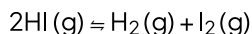
Calculate the total pressure of the reaction, if $K_p = 9.8 \times 10^{-3} \text{ kPa}$.

$P = \dots\dots\dots \text{ kPa}$

[4 marks]

Question 4c

Hydrogen iodide can undergo thermal decomposition according to the following equation:



The value of K_p for this reaction at 700 K is 0.0185.

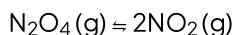
Calculate the total pressure when the partial pressure of hydrogen iodide, at equilibrium, is 133 kPa. Give your answer to the appropriate number of significant figures.

$P = \dots\dots\dots$

[4 marks]

Question 4d

Dinitrogen tetroxide decomposes according to the following equilibrium:



The value of K_p for this reaction at 300 K is 385 kPa.

The equilibrium mixture contained 2.40 moles of nitrogen dioxide after an initial 3.00 moles of dinitrogen tetroxide were heated at 300 K

Calculate the partial pressures of dinitrogen tetroxide and nitrogen dioxide. Give your answer to the appropriate number of significant figures.

$p\text{ N}_2\text{O}_4 = \dots\dots\dots$ kPa

$p\text{ NO}_2 = \dots\dots\dots$ kPa

[6 marks]**Question 5a**

Malonic acid is a weak dibasic carboxylic acid with the formula $C_3H_4O_4$. Draw the displayed structure of malonic acid.

[1 mark]**Question 5b**

Suggest, with a reason, whether ethanoic acid or malonic acid has a higher pH.

[2 marks]**Question 5c**

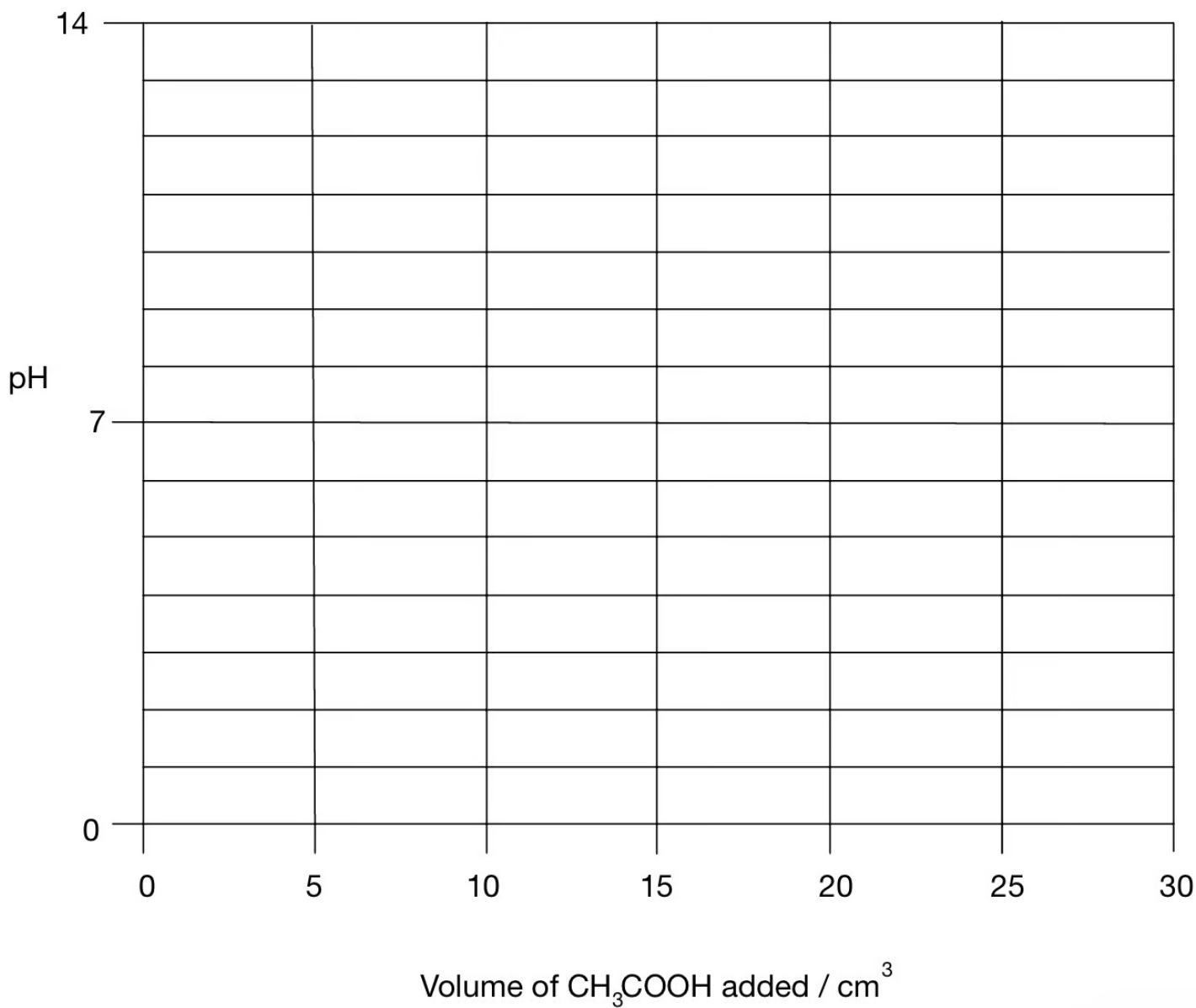
Apart from testing the pH, suggest how equimolar solutions of malonic acid and ethanoic acid may be distinguished.

[1 mark]

Question 5d

30.0 of $0.075 \text{ mol dm}^{-3}$ CH_3COOH with a pH of 2.93 was slowly added to a sample of 10.0 cm^3 $0.150 \text{ mol dm}^{-3}$ KOH with a pH of 13.1. The pH was measured throughout the addition.

On the following axes, sketch how the pH will change during the addition of a total of 30.0 of $0.075 \text{ CH}_3\text{COOH}$.



[3 marks]

Model Answers: Hard

1a

a)

i) The definition of dynamic equilibrium is:

- Forward and backward reactions proceeding at an equal rate; [1 mark]
- The concentration / (number of) moles of reactants and products remain constant; [1 mark]

ii) The expression for K_c for this reaction is:

- $K_c = \frac{[\text{C}]^2}{[\text{A}]^3[\text{B}]^2}$; [1 mark]

[Total: 3 marks]

- The definition of dynamic equilibrium is an important definition to learn
- **Remember:** When writing the expression $K_c = \frac{[\text{products}]}{[\text{reactants}]}$
 - Use [] to represent concentration
 - If you use () you will not be awarded the mark

1b

b) To calculate the concentration of **B** in the equilibrium mixture if the volume of the container is 8.00 dm³:

- Correct rearranged equation

$$[\text{B}]^2 = \frac{[\text{C}]^2}{[\text{A}]^3 \times K_c}; [1 \text{ mark}]$$

- Correct dividing by volume in dm³

$$[\text{B}]^2 = \frac{(4.28/8)^2}{(3.34/8)^3 \times 255}; [1 \text{ mark}]$$

- Correct calculation of $[B]^2$
 $[B]^2 = 0.0154$; [1 mark]
- Correct use of $\sqrt{}$
 $[B] = 0.124$ to 3 significant figures; [1 mark]

[Total: 4 marks]

- The first step here is to rearrange the equation in order to calculate $[B]^2$
- Then, you must ensure that you divide the moles values given by the volume of the container
 - The concentration depends on the volume
- You will then have a value for $[B]^2$ so can then take the square root of this value to give the concentration of **B**

1c

c) The effect, if any, on the concentration of **B** at equilibrium if temperature **T** is increased is:

- (The concentration of **B**) increases; [1 mark]

Explanation:

- Increasing temperature favours the endothermic / reverse pathway / reaction; [1 mark]

[Total: 2 marks]

- Le Chatelier's principle states that if a change is made to a system then the system will oppose this change
 - By decreasing the temperature the system will favour the exothermic pathway
 - By increasing the temperature the system will favour the endothermic pathway
- $3A(g) + 2B(g) \rightleftharpoons 2C(g) \quad \Delta H = -65 \text{ kJ mol}^{-1}$
- The forward reaction has a negative ΔH value so it is exothermic
- Therefore, increasing the temperature favours the reverse reaction which means that there will be an increased concentration of **B**

1d

d) Calculate the equilibrium constant, K_c , for the following reaction at temperature, T :

- $\frac{1}{255} = 0.00392$; [1 mark]

[Total: 1 mark]

- The reaction in part (d) is the reverse of the reaction in part (a)
- The K_c of the reverse reaction is related to the K_c of the forward reaction because the two values are reciprocals of one another
- You can think of this as the products and reactants for the K_c of the reverse reaction are flipped compared to the forward reaction
- If the K_c of the forward reaction is 255 then the K_c of the reverse reaction is $\frac{1}{255}$

2a

a) The forward reaction is:

- Exothermic; [1 mark]
- As temperature increases, the % yield decreases; [1 mark]
- The equilibrium shifts to the left-hand side / favours backward reaction to oppose the change in temperature / increase in temperature; [1 mark]

[Total: 3 marks]

- The graph in Fig 2.1 shows as the temperature is increasing the yield of the product is decreasing at any given pressure
- Le Chatelier's principle states
 - If a change is made to a dynamic equilibrium by changing the conditions, the position of equilibrium moves to counteract the change
- In this case, we can see that the forward reaction must be exothermic
 - An increase in temperature causes the yield to decrease, therefore a higher temperature favours the backward reaction
 - A decrease in temperature causes the yield to increase, therefore a lower temperature favours the forward reaction
- An exothermic pathway is favoured by lowering the temperature
- An endothermic pathway is favoured by increasing the temperature

2b

b) Explanation of whether the forward reaction will involve either an increase or decrease in

the number of moles of a gas:

- Increase in number of moles of gas; [1 mark]
- Lower pressure gives greater yield; [1 mark]
- Equilibrium shifts to LHS / backward reaction favoured by higher pressure as fewer moles / molecules of gas on LHS of the equation

OR

Equilibrium shifts to RHS / forward reaction favoured by lower pressure as more moles / molecules of gas on RHS of the equation; [1 mark]

[Total: 3 marks]

- The graph in Fig 2.1 shows that as the pressure decreases, the yield of the product increases
- Le Chatelier's principle states
 - If a change is made to a dynamic equilibrium by changing the conditions, the position of equilibrium moves to counteract the change
- There are three different pressures shown in Fig 2.1; 5MPa, 20MPa and 30MPa
 - The highest yield is produced when the pressure is 5MPa
 - An increase in pressure causes the yield to decrease, therefore a higher pressure favours the backward reaction
 - A decrease in pressure causes the yield to increase, therefore a lower pressure favours the forward reaction
- Higher pressure will shift the equilibrium to the side with the fewest gaseous moles
- Lower pressure will shift the equilibrium to the side with the most gaseous moles

2c

c) The **forward reaction** is:

- Exothermic; [1 mark]

Explanation:

- As the temperature increases, K_c , decreases; [1 mark]
- Yield decreases; [1 mark]

[Total: 3 marks]

- The value for K_c can only be changed by a difference in temperature as it is a constant
- $K_c = \frac{[\text{products}]}{[\text{reactants}]}$
 - If the value of K_c decreases, the concentration of reactants will be increased
 - If the value of K_c increases, the concentration of the products will increase
- For an exothermic reaction
 - To increase the concentration of reactants, you increase the temperature
 - This will decrease the value of K_c
 - As the bottom line of the equation will have a larger value
- For an endothermic reaction
 - To increase the concentration of reactants, you decrease the temperature
 - This will decrease the value of K_c
 - As the bottom line of the equation will have a larger value

$$K_c = \frac{[\text{products}]}{[\text{reactants}]}$$

If this value increases, K_c increases

If this value increases, K_c decreases

2d

d) A chemist needs to know the value of K_c , at a given temperature because:

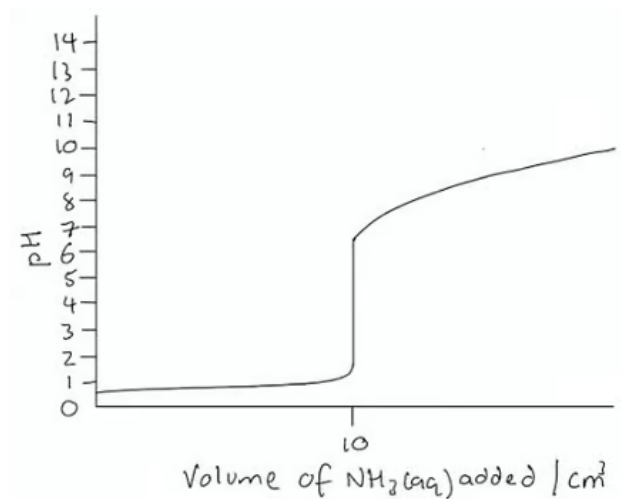
- If K_c is low, the yield of the product is small; [1 mark]
- May **not** be cost-effective to produce low yield at a high temperature; [1 mark]

[Total: 2 marks]

- It is important to consider the % of yield produced at a given temperature when producing a particular product
 - If K_c is low, it will mean that the % of product produced is low
 - If K_c is high, it will mean that the % of product produced is high
- If the reaction is endothermic and a very high temperature is required for the production of the desired product then it may not be a cost-effective procedure

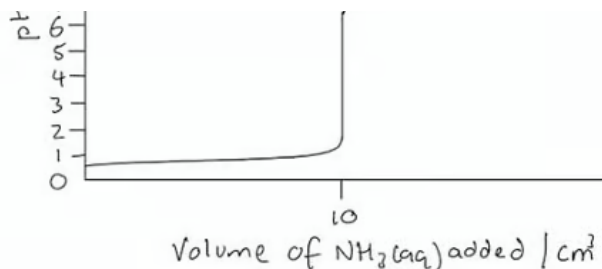
3a

a) The pH curve that would be obtained from this titration is:



- Initial pH between pH 0 and 2; [1 mark]
- Vertical line at 10 cm^3 ; [1 mark]
- Line tails off between pH 9 and 11; [1 mark]

[Total: 3 marks]



- Initial pH between pH 0 and 2; [1 mark]
- Vertical line at 10 cm³; [1 mark]
- Line tails off between pH 9 and 11; [1 mark]

[Total: 3 marks]

- pH curves are easy to get wrong when drawing them from scratch
- You have been given the volume of hydrochloric acid so can deduce that this is in the conical flask
 - The ammonia is therefore being added from the burette so the volume of ammonia goes on the x-axis
- HCl is a strong acid so your curve must begin at less than pH 1
- Ammonia is a weak base so the curve should tail off at approximately pH 10
- The concentrations of ammonia and hydrochloric acid are the same as one another
 - So they will react in equal quantities causing the pH will change at 10.0 cm³
- This is a strong acid-weak base titration curve so the midpoint of the curve (the equivalence point) should be approximately pH 5

3b

b) Methyl red is a suitable indicator for this titration but phenolphthalein is not because:

- Methyl red changes colour at the equivalence point; [1 mark]
- Phenolphthalein changes colour between pH 8 - 10 which does not correspond to the steepest part of the graph; [1 mark]

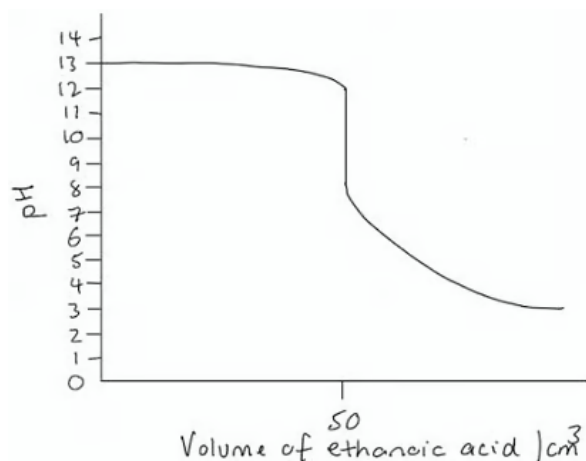
[Total: 2 marks]

- The pH range for methyl red is 4.2 - 6.3 which corresponds to the values where the sharpest pH change occurs
 - From the graph, this is between pH 3.0 and pH 7.5
- The pH range for phenolphthalein does not correspond to the region where there is the

sharpest pH change

3c

c) The graph that would be obtained for this titration is:



- Initial pH at approximately between 12 and 14; [1 mark]
- Vertical line at 50 cm³; [1 mark]
- Line tails off between pH 3 and 5; [1 mark]

[Total: 3 marks]

- The sodium hydroxide will be in the conical flask and ethanoic acid will be added
 - So, the volume of ethanoic acid needs to be on the x-axis
- Sodium hydroxide is a strong base so the initial pH needs to be high
- Ethanoic acid is a weak acid so the curve should tail off at approximately pH 3
- The concentration of sodium hydroxide (0.200 mol dm⁻³) is double that of the ethanoic acid (0.100 mol dm⁻³) but the molar ratio is equal which means that double the volume of ethanoic acid is required for neutralisation to occur
 - The sharpest pH change should, therefore, be at 50 cm³ as 25.0 cm³ sodium hydroxide was used

3d

d) Phenolphthalein is a suitable indicator for this titration but methyl red is not because::

- Phenolphthalein changes colour at the equivalence point / where the sharpest pH change is; [1 mark]

- Methyl red changes colour between pH 3 and 4.5 (which does not correspond to the steepest pH change); [1 mark]

[Total: 2 marks]

- The pH range for phenolphthalein is 8 - 10 which corresponds to the values where the sharpest pH change occurs
- The pH range for methyl red is 4.2 - 6.3 which does not correspond to the values where the sharpest pH change occurs
 - From the graph, this is between approximately pH 7.5 and pH 11.0

3e

e) Explaining whether the other two indicators in Table 3.1 are suitable for use in either titration:

- Bromocresol green is suitable for the HCl / NH₃ titration; [1 mark]
- (Because) its pH range covers that of the equivalence point

OR

The pH range corresponds to where the equivalence point lies; [1 mark]

- Neither indicator is suitable for the CH₃COOH / NaOH titration; [1 mark]
- (Because) neither of their pH ranges corresponds to where the steepest pH change occurs

OR

Neither corresponds to where the equivalence point lies; [1 mark]

[Total: 4 marks]

- Similar to questions (b) and (d), you just need to check the pH range against the steepest pH change on your titration curves

4a

a) To calculate the total pressure of the reaction:

	Step	Calculations
1	Calculate the total moles	$0.192 + 0.822 + 0.822 = 1.836$ moles; [1 mark]

2	Calculate the mole fractions	$PCl_5 = \frac{0.192}{1.836} = 0.104575$ $\text{AND } PCl_3 = \frac{0.822}{1.836} = 0.44771$ $\text{AND } Cl_2 = \frac{0.822}{1.836} = 0.44771; [1 \text{ mark}]$
3	Write the K_p expression	$K_p = \frac{p_{PCl_3} (g) \text{ eqm} \times p_{Cl_2} (g) \text{ eqm}}{p_{PCl_5} (g) \text{ eqm}}; [1 \text{ mark}]$
4	Substitute values and evaluate (Assume that total pressure = P)	$526 = \frac{(0.44771P) \times (0.44771P)}{(0.104575P)}$ $526 = \frac{0.44771^2 \times P}{0.104575}$ $P = \frac{55.01}{0.200}$ $P = 275 \text{ kPa}; [1 \text{ mark}]$

[Total: 4 marks]

- This question follows the standard K_p calculation process to start with, calculating total moles and using equilibrium moles to calculate mole fractions
- Then the K_p expression is written
- At this point, you are unable to calculate the partial pressures because you do not have the total pressure
- So, you have to use algebraic/mathematical methods to lead you to the final answer, using P to represent the total pressure that you are trying to find
- **Remember:** The question states that the answer must be given to the appropriate number of significant figures, which in this case is 3 significant figures

4b

b) To calculate the total pressure of the reaction:

	Step	Calculation
1	Calculate the total moles	$0.92 + 0.06 + 0.04 = 1.02 \text{ moles}; [1 \text{ mark}]$

2	Calculate the mole fractions	$\text{NO}_2 = \frac{0.92}{1.02} = 0.901961$ $\text{AND NO} = \frac{0.06}{1.02} = 0.058824$ $\text{AND O}_2 = \frac{0.04}{1.02} = 0.039216; [1 \text{ mark}]$
3	Write the K_p expression	$K_p = \frac{p^2\text{NO (g) eqm} \times p\text{O}_2 \text{ (g) eqm}}{p^2\text{NO}_2 \text{ (g) eqm}}; [1 \text{ mark}]$
4	Substitute values and evaluate	$9.8 \times 10^{-3} = \frac{(0.058824P)^2 \times (0.039216P)}{(0.901961P)^2}$ $9.8 \times 10^{-3} = \frac{3.460263 \times 10^{-3} P^2 \times 0.039216P}{0.81353 P^2}$ $9.8 \times 10^{-3} = \frac{3.460263 \times 10^{-3} \times 0.039216P}{0.81353}$ $9.8 \times 10^{-3} = 1.668 \times 10^{-4} \times P$ $P = \frac{9.8 \times 10^{-3}}{1.668 \times 10^{-4}}$ $P \approx 58.8 \text{ kPa}; [1 \text{ mark}]$

[Total: 4 marks]

- Like part (a), this question follows the standard K_p calculation process but then has to switch to an algebraic / mathematical methods to lead you to the final answer
- The algebra / maths to get the total pressure is more complex as the K_p expression automatically has terms that are squared and that can cancel out

4c

c) To calculate the total pressure when the partial pressure of hydrogen iodide, at equilibrium, is 133 kPa:

	Step	Calculation
		$K_p =$

1	Write the K_p expression	$\frac{p_{\text{H}_2(\text{g}) \text{ eqm}} \times p_{\text{I}_2(\text{g}) \text{ eqm}}}{p_{\text{HI}(\text{g}) \text{ eqm}}^2}$; [1 mark]
2	Recognise that hydrogen and iodine are produced in equal amounts	$K_p = \frac{p_{\text{H}_2(\text{g}) \text{ eqm}}}{p_{\text{HI}(\text{g}) \text{ eqm}}^2} \text{ OR } \frac{p_{\text{I}_2(\text{g}) \text{ eqm}}}{p_{\text{HI}(\text{g}) \text{ eqm}}^2}$; [1 mark]
3	Substitute values and evaluate	$0.0185 = \frac{p_{\text{H}_2(\text{g}) \text{ eqm}}}{133^2}$ $p_{\text{H}_2(\text{g}) \text{ eqm}} = 0.0185 \times 133^2$ $p_{\text{H}_2(\text{g}) \text{ eqm}} = \sqrt{(0.0185 \times 133^2)}$

		$\text{AND } O_2 = \frac{0.04}{1.02} = 0.039216; [1 \text{ mark}]$
3	Write the K_p expression	$K_p = \frac{p^2NO(g)_{eqm} \times pO_2(g)_{eqm}}{p^2NO_2(g)_{eqm}}; [1 \text{ mark}]$
4	Substitute values and evaluate	$9.8 \times 10^{-3} = \frac{(0.058824P)^2 \times (0.039216P)}{(0.901961P)^2}$ $9.8 \times 10^{-3} = \frac{3.460263 \times 10^{-3} P^2 \times 0.039216P}{0.81353 P^2}$ $9.8 \times 10^{-3} = \frac{3.460263 \times 10^{-3} \times 0.039216P}{0.81353}$ $9.8 \times 10^{-3} = 1.668 \times 10^{-4} \times P$ $P = \frac{9.8 \times 10^{-3}}{1.668 \times 10^{-4}}$ $P = 58.8 \text{ kPa}; [1 \text{ mark}]$

[Total: 4 marks]

- Like part (a), this question follows the standard K_p calculation process but then has to switch to an algebraic / mathematical methods to lead you to the final answer
- The algebra / maths to get the total pressure is more complex as the K_p expression automatically has terms that are squared and that can cancel out

4c

c) To calculate the total pressure when the partial pressure of hydrogen iodide, at equilibrium, is 133 kPa:

	Step	Calculation
1	Write the K_p expression	$K_p = \frac{pH_2(g)_{eqm} \times pI_2(g)_{eqm}}{p^2 HI(g)_{eqm}}; [1 \text{ mark}]$

2	Recognise that hydrogen and iodine are produced in equal amounts	$K_p = \frac{p^2\text{H}_2(\text{g}) \text{ eqm}}{p^2\text{HI}(\text{g}) \text{ eqm}} \quad \text{OR}$ $\frac{p^2\text{I}_2(\text{g}) \text{ eqm}}{p^2\text{HI}(\text{g}) \text{ eqm}} ; [1 \text{ mark}]$
3	Substitute values and evaluate	$0.0185 = \frac{p^2\text{H}_2(\text{g}) \text{ eqm}}{133^2}$ $p^2\text{H}_2(\text{g}) \text{ eqm} = 0.0185 \times 133^2$ $p^2\text{H}_2(\text{g}) \text{ eqm} = \sqrt{(0.0185 \times 133^2)}$ $p\text{H}_2(\text{g}) \text{ eqm} = 18.1 \text{ to } 3 \text{ s.f.}$ AND Therefore, $p\text{I}_2(\text{g}) \text{ eqm} = 18.1$ to 3 s.f.; [1 mark]
4	Calculate the total pressure	$18.1 + 18.1 + 133 = 169 \text{ kPa to } 3 \text{ s.f.}; [1 \text{ mark}]$

[Total: 4 marks]

- Unlike parts (a) and (b), this question doesn't require the algebraic / mathematical approach
- However, it does require you to recognise that the hydrogen iodide will decompose to give equal amounts of hydrogen and iodine.
- This means that the partial pressures of hydrogen and iodine will be the same
- This also means that solving the K_p expression is more simplistic than you may think
 - Basically, rearrange the expression
- **Remember:** Perform the square root of your rearranged expression to get the final answer - this step is often missed out in exams

4d

d) To calculate the partial pressures of dinitrogen tetroxide and nitrogen dioxide:

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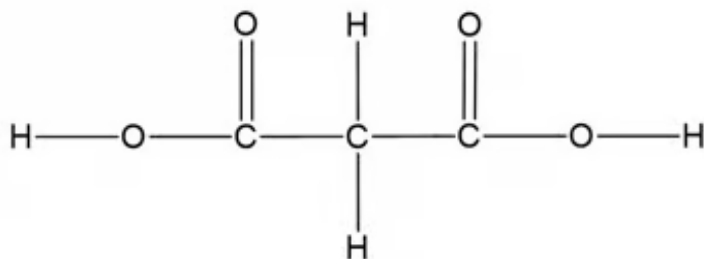
	Steps	Gas	Initial moles	Change	Equilibrium moles
1	Calculate the equilibrium moles of N_2O_4	N_2O_4	3.00	-1.20	1.80; [1 mark]
		NO_2	0.00	+2.40	2.40
2	Calculate the total moles and the mole fractions	Total moles = $1.8 + 2.4 = 4.2$ moles AND $\text{N}_2\text{O}_4 = \frac{1.8}{4.2} = 0.42857$ AND $\text{NO}_2 = \frac{2.4}{4.2} = 0.571428$; [1 mark]			
3	Write the K_p expression	$K_p = \frac{p^2 \text{NO}_2 (\text{g}) \text{ eqm}}{p \text{N}_2\text{O}_4 (\text{g}) \text{ eqm}}$; [1 mark]			
4	Express the partial pressures and substitute values in the K_p expression	$385 = \frac{(0.571428 \times P)^2}{(0.42857 \times P)}$; [1 mark]			
5	Evaluate the K_p expression to calculate the total pressure, P	$385 = \frac{(0.571428)^2 \times P}{0.42857}$ $P = \frac{(385 \times 0.42857)}{(0.571428)^2}$ $P = 505.31 \text{ kPa}$; [1 mark]			
6	Calculate the partial pressures	$\text{N}_2\text{O}_4 = 0.42857 \times 505.31 = 216.56 \text{ kPa}$ AND $\text{NO}_2 = 0.571428 \times 505.31 = 288.75 \text{ kPa}$; [1 mark]			

[Total: 6 marks]

- This question goes one step beyond parts (a) and (b) by asking you to calculate the partial pressures
- To do this, simply put the total pressure that you calculate back into the expressions that you use in your algebraic / mathematical expressions that you used

5a

a) The displayed structure of malonic acid is:



- Correct diagram showing all the bonds; [1 mark]

[Total: 1 mark]

- Carboxylic means malonic acid must contain the -COOH group
- Dibasic means it reacts with two mole equivalents of base, so it must contain two -COOH groups

5b

b) Whether ethanoic or malonic acid has the higher pH:

- Ethanoic acid; [1 mark]
- Malonic acid is diprotic (and produces more hydrogen ions than ethanoic acid)
OR
Ethanoic acid is monoprotic (and produces fewer hydrogen ions than ethanoic acid); [1 mark]

[Total: 2 marks]

- **Careful:** The question asks which acid (with the same concentration) will have the highest pH, which means which is the weaker acid
 - Ethanoic acid is the weaker acid so will have a higher pH as it will release fewer hydrogen ions

5c

c) Equimolar solutions of malonic acid and ethanoic acid may be distinguished by:

Any **one** from

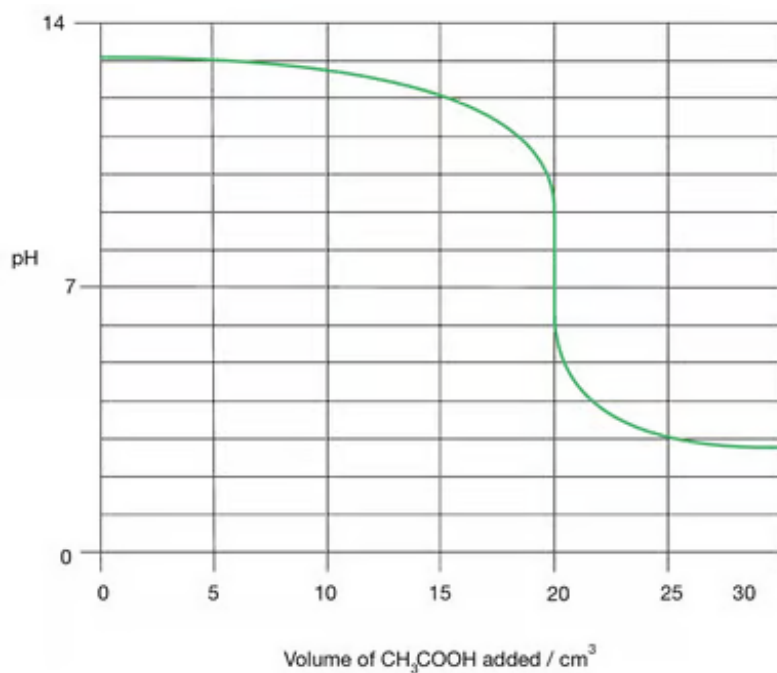
- Test the conductivity of the solutions
AND
malonic acid should show higher conductivity; [1 mark]
- Add a reactive metal, e.g. Mg
AND
malonic acid should produce bubbles slightly faster; [1 mark]
- Add a metal oxide/carbonate
AND
the rate of reaction should be (slightly) faster with malonic acid; [1 mark]

[Total: 1 mark]

- Equimolar means equal concentration; unless the solutions are equimolar their relative acidity cannot be compared

5d

d) The pH titration curve should be:



- Starting pH = 13.1; [1 mark]
- Vertical section at 20.0 cm^3

AND

Middle of the vertical section above pH 7; [1 mark]

- Final pH = 2.93; [1 mark]

[Total: 3 marks]

- To work out the volume of ethanoic acid required, use:
 - $\text{Vol}_a \times \text{Conc}_a = \text{Vol}_b \times \text{Conc}_b$
 - $\text{Vol}_a \times 0.075 = 0.150 \times 10$
 - $\text{Vol}_a = \frac{0.150 \times 10}{0.075} = 20.0 \text{ cm}^3$
- As the acid is being added to the base, the pH must start high and then end low
 - Pay close attention to which way round the titration is being carried out
- The middle of the vertical section must be above pH 7 due to the fact that a weak acid is being added to a strong base

